Quantum Optimization of Currency Arbitrage via Graph-Informed Entanglement Strategies

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Abstract

Detecting arbitrage opportunities in the foreign exchange market is computationally challenging, as the problem's NP-completeness causes complexity to grow rapidly with the size of the currency network. While previous studies often relied on simplified models that overlook real-world trading frictions & microstructures, this paper explicitly incorporates the real-world elements: bid and ask rates, into the QUBO formulation and evaluates performance under QAOA with various entanglement strategies. Results show that a modular, graph-informed entanglement structure achieves near-optimal approximation ratios (approximately 0.999). By integrating market spread dynamics into the quantum optimization process, this work advances beyond existing arbitrage models and demonstrates the broader applicability of graph-based quantum algorithms for financial optimization.

CCS Concepts

• **Theory of computation → Quantum computation theory**; • **Mathematics of computing → Combinatorial optimization**; • **Social and professional topics → Computing and business**.

Keywords

Quantum Optimization, Currency Arbitrage, Quantum Computation Theory, Combinatorial Optimization

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1 Introduction

Arbitrage — the practice of profiting from price differences of an asset — is a key mechanism in the foreign exchange (FX) market, the world's largest and most liquid financial market with over $6.6 trillion traded daily [19]. Arbitrage opportunities arise due to the market's vast scale and volatility, and can be modeled as a graph-based optimization problem, with currencies as nodes and exchange rates as edges. Traditional methods like the Bellman–Ford algorithm or linear programming, while effective for smaller networks, struggle to scale due to the exponential growth of the search space and the NP-completeness of arbitrage detection [3]. Quantum optimization techniques, however, offer a promising approach to address this computational challenge [1].

Quantum computing offers new possibilities for solving combinatorial problems in finance by leveraging quantum properties such as superposition and entanglement to explore large solution spaces efficiently [16]. One prominent example is the Quantum Approximate Optimization Algorithm (QAOA) [7], which has shown promise in areas such as portfolio optimization [2] and may improve arbitrage detection at scale. However, QAOA's performance on Noisy Intermediate-Scale Quantum (NISQ) devices [18] depends heavily on the chosen entanglement strategy, since noise can limit coherence time and solution quality [24].

This work integrates bid and ask rates into the problem formulation for more realistic arbitrage modeling and evaluate multiple entanglement strategies under noise. The results provide insights into the trade-offs between entanglement design and optimization quality for FX arbitrage problems.

2 Background

2.1 Currency Arbitrage

The vast size of the FX market — trading trillions of dollars daily as of 2017 [9] — offers numerous currency arbitrage opportunities. A classic example is triangular arbitrage, which involves three currencies and their exchange rates. For instance, starting with USD, one could convert USD to EUR, then EUR to GBP, and finally GBP back to USD. If the final amount exceeds the initial, an arbitrage opportunity exists. However, factors like transaction costs and the choice of intermediate currencies can impact profitability [8]. Mathematically, triangular arbitrage is evaluated as the product of exchange rates: if the product of the exchange rates in a cycle (e.g., USD → EUR → GBP → USD) exceeds 1, arbitrage exists.

2.2 Classical Arbitrage Models

Arbitrage opportunities can be identified using classical methods such as the Bellman–Ford algorithm [23]. Originally designed to find shortest paths in graphs with negative edge weights, it detects arbitrage cycles when exchange rates are expressed as log-transformed weights. Despite its theoretical effectiveness, its practical use is limited by market frictions, transaction costs, and rapidly

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changing rates [22]. Moreover, its time complexity of $O(|V||E|)$ becomes prohibitive on large graphs [5].

Alternatively, arbitrage can be formulated as a combinatorial optimization problem via binary integer programming (BIP) [21]:

$$\begin{aligned} \max \quad & \sum_{i} \sum_{j \neq i} \log(r_{ij})\, x_{ij}, \\ \text{s.t.} \quad & \sum_{j \neq i} x_{ji} - \sum_{j \neq i} x_{ij} = 0 \quad \text{for } i = 1, \ldots, m, \\ & x_{ij} \in \{0, 1\} \qquad\qquad \text{for } i, j = 1, \ldots, m,\ j \neq i. \end{aligned} \tag{1}$$

Here, r_{ij} is the exchange rate from currency i to j, and $x_{ij} \in \{0, 1\}$ indicates whether that trade is included in the arbitrage cycle. The original objective maximizes the product of exchange rates along a cycle; taking logarithms reduces computational intensity by taking on an additive form:

$$\max \ \log\Big(\prod_{(i,j)} r_{ij}^{x_{ij}}\Big) \ = \ \max \ \sum_{(i,j)} x_{ij} \ \log(r_{ij}). \tag{2}$$

This formulation enforces flow conservation at each node and binary selection of trades, ensuring that the cycle is closed and each trade is either used or not.

2.3 Quantum Arbitrage Models

Quantum computing offers new promise for solving financial optimization problems more efficiently than classical approaches [12]. In particular, quantum models for arbitrage detection can improve scalability and solution quality compared to classical methods [20].

2.3.1 QUBO Model Basics. Quantum optimization typically begins by formulating the problem as a Quadratic Unconstrained Binary Optimization (QUBO):

$$\min_{x \in \{0,1\}^n} \ x^\top Q\, x, \tag{3}$$

where x is a binary vector and Q is a real symmetric matrix of coefficients. QUBO problems map directly to the Ising model, making them well suited for implementation on quantum hardware [10].

2.3.2 Gate-Based Quantum Computing & the Quantum Approximate Optimization Algorithm (QAOA). Gate-based quantum computing uses qubits and quantum gates to evolve quantum states analogously to classical circuits. Algorithms are expressed as sequences of unitary operations that steer the state toward high-quality solutions.

QAOA, introduced by Farhi *et al.* [7], is a variational quantum algorithm for combinatorial optimization. It alternates between a cost Hamiltonian H_C, which encodes the objective function, and a mixing Hamiltonian H_M, which promotes exploration. After p layers, the state is

$$\big|\psi(\boldsymbol{\gamma}, \boldsymbol{\beta})\big\rangle = \prod_{k=1}^{p} e^{-i\beta_k H_M}\, e^{-i\gamma_k H_C}\, \big|s\big\rangle, \tag{4}$$

where $|s\rangle$ is typically the uniform superposition state, and $\boldsymbol{\gamma}, \boldsymbol{\beta}$ are the layer-specific variational parameters.

QAOA performance depends critically on the structure of H_C, the circuit depth p, and the entanglement pattern among qubits. Deeper circuits can achieve better approximations but require more gates and are more susceptible to noise. This work focuses on how various entanglement strategies interact with the complexity of this bid–ask QUBO formulation to influence circuit depth, gate count, and ultimately solution quality under realistic noise.

3 Formulation

This section formulates the currency arbitrage problem with bid & ask rates, capturing real-world asymmetries in exchange rates and enabling QUBO-based optimization via QAOA.

3.1 Linear Programming Formulation

Exchange rates in FX markets are typically quoted with two prices: the bid (the price at which dealers buy a currency) and the ask (the price at which they sell it). The difference, known as the bid–ask spread, represents transaction costs and dealer profit, influencing both arbitrage feasibility and profitability.

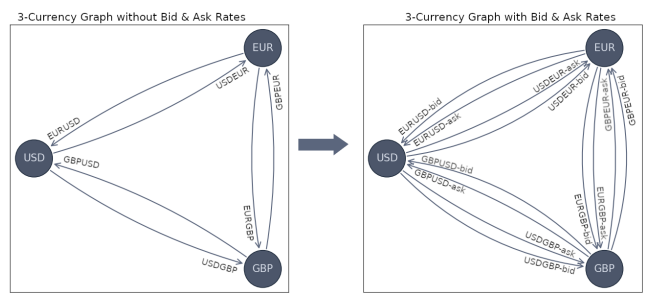

Figure 1: Fully connected, bidirectional currency graph with bid and ask rates. Each trade direction now has two edges, doubling both the edge count and the required qubits.

Incorporating bid and ask rates introduces market realism by accounting for transaction costs and liquidity, but also increases model complexity. Many prior models simplify arbitrage detection by assuming symmetric or single-rate conversions.

3.1.1 Asymmetric Bid and Ask Rates. Each currency pair includes a bid rate r_{ij}^{bid} and an ask rate r_{ij}^{ask}, corresponding to the market maker's buying and selling prices. The spread $\big(r_{ij}^{\text{ask}} - r_{ij}^{\text{bid}}\big)$ reflects costs, liquidity, and other market factors.

Asymmetry complicates optimization: selling currency i converts at the bid rate, while buying currency i uses the ask rate. Consequently, each currency pair introduces two decision variables, increasing the problem's dimensionality.

3.2 Graph Representation of the Currency Network

The currency network is modeled as a fully connected, bidirectional multidigraph $G = (N, E)$, where N denotes the set of currencies and E denotes the set of directed exchange edges. Each ordered pair (i, j) includes:

- **Bid Edge**: selling currency i at rate r_{ij}^{bid} to obtain currency j.
- **Ask Edge**: buying currency i at rate r_{ij}^{ask} using currency j.

3.3 Binary Integer Programming Formulation

Arbitrage detection is formulated as a binary integer program using decision variables $x_{ij}^q \in \{0,1\}$ for each directed edge (i,j) and transaction type $q \in \{\text{bid}, \text{ask}\}$:

$$x_{ij}^q = \begin{cases} 1, & \text{if the transaction from } i \text{ to } j \text{ at rate } r_{ij}^q \text{ is chosen,} \\ 0, & \text{otherwise.} \end{cases} \tag{5}$$

3.3.1 Objective Function. The objective maximizes the total logarithmic return of selected exchanges. Taking logarithms linearizes the multiplicative structure of currency conversions:

$$\max \sum_{i,j} \sum_{q \in \{\text{bid},\text{ask}\}} (1-\delta_{ij})\, x_{ij}^q \, \log(r_{ij}^q), \tag{6}$$

where:

- r_{ij}^q is the bid or ask rate from currency i to j.
- δ_{ij} is the Kronecker delta, used to exclude self-loops.

3.3.2 Constraints. The following constraints ensure that only valid arbitrage cycles are found:

Flow Conservation. For each currency $i \in N$, net inflow equals outflow, ensuring a closed cycle:

$$\sum_j \sum_q (1-\delta_{ij})\, x_{ji}^q - \sum_k \sum_q (1-\delta_{ik})\, x_{ik}^q = 0, \quad \text{for } i \in N. \tag{7}$$

Inflow and Outflow. Each currency may have at most one incoming and one outgoing exchange:

$$\sum_j \sum_q (1-\delta_{ij})\, x_{ij}^q \le 1, \quad \text{for } i \in N, \tag{8a}$$

$$\sum_j \sum_q (1-\delta_{ij})\, x_{ji}^q \le 1, \quad \text{for } i \in N. \tag{8b}$$

Binary Variable Constraint: All decision variables are binary.

$$x_{ij}^q \in \{0,1\}, \quad \text{for } (i,j) \in E, \quad q \in \{\text{bid}, \text{ask}\}. \tag{9}$$

3.4 Representing Asymmetric Bid and Ask Rates

Decision variables x_{ij}^{bid} and x_{ij}^{ask} distinguish transactions at bid and ask rates, respectively. Selling currency i for j (bid) differs from buying j using i (ask).

Impact on the Objective Function. Each chosen transaction contributes:

$$\log(r_{ij}^q) = \begin{cases} \log(r_{ij}^{\text{bid}}), & q = \text{bid}, \\ \log(1/r_{ij}^{\text{ask}}) = -\log(r_{ij}^{\text{ask}}), & q = \text{ask}. \end{cases} \tag{10}$$

3.5 Complete Formulation

The full BIP with asymmetric bid and ask rates is defined as follows:

Objective Function.

$$\max \sum_{i,j} (1-\delta_{ij}) \left[x_{ij}^{\text{bid}} \log r_{ij}^{\text{bid}} - x_{ij}^{\text{ask}} \log r_{ij}^{\text{ask}} \right]. \tag{11}$$

Constraints.

(1) Flow Conservation Constraint:

$$\sum_j (1-\delta_{ij})(x_{ji}^{\text{bid}} + x_{ji}^{\text{ask}}) - \sum_k (1-\delta_{ik})(x_{ik}^{\text{bid}} + x_{ik}^{\text{ask}}) = 0, \quad \text{for } i \in N. \tag{12}$$

(2) Inflow and Outflow Constraints:

$$\sum_j (1-\delta_{ij})(x_{ij}^{\text{bid}} + x_{ij}^{\text{ask}}) \le 1, \quad \text{for } i \in N, \tag{13a}$$

$$\sum_j (1-\delta_{ij})(x_{ji}^{\text{bid}} + x_{ji}^{\text{ask}}) \le 1, \quad \text{for } i \in N. \tag{13b}$$

(3) Binary Variable Constraint:

$$x_{ij}^q \in \{0,1\}, \quad \text{for } (i,j) \in E, \; q \in \{\text{bid}, \text{ask}\}. \tag{14}$$

3.6 QUBO Conversion

To enable quantum optimization, the BIP formulation is transformed into a QUBO problem. QUBO expresses the objective as a quadratic function of binary variables, absorbing constraints through penalty terms. This structure is compatible with quantum algorithms such as QAOA. The standard QUBO form is:

$$\min_{x \in \{0,1\}^n} x^\top Q\, x, \tag{15}$$

where Q is a real symmetric matrix encoding the problem's coefficients, and x is the vector of binary decision variables.

Objective Function Transformation. The original objective is recast as a minimization problem:

$$\min \left(-\sum_{i,j} (1-\delta_{ij}) \left[x_{ij}^{\text{bid}} \log r_{ij}^{\text{bid}} + x_{ij}^{\text{ask}} \log \left(\frac{1}{r_{ji}^{\text{ask}}} \right) \right] \right). \tag{16}$$

3.6.1 Incorporating Constraints as Penalty Terms. To remove explicit constraints, penalty terms are added to the objective function. Each constraint is weighted by a positive penalty coefficient λ.

Flow Conservation Penalty. For each currency $i \in N$:

$$\sum_j (1-\delta_{ij})(x_{ji}^{\text{bid}} + x_{ji}^{\text{ask}}) - \sum_k (1-\delta_{ik})(x_{ik}^{\text{bid}} + x_{ik}^{\text{ask}}) = 0. \tag{17}$$

the corresponding penalty term is:

$$P_{\text{flow}} = \lambda_{\text{flow}} \sum_i \left(\sum_j (1-\delta_{ij})(x_{ji}^{\text{bid}} + x_{ji}^{\text{ask}}) - \sum_k (1-\delta_{ik})(x_{ik}^{\text{bid}} + x_{ik}^{\text{ask}}) \right)^2 \tag{18}$$

Inflow and Outflow Penalties. For each currency $i \in N$:

$$\sum_j (1-\delta_{ij})(x_{ij}^{\text{bid}} + x_{ij}^{\text{ask}}) \leq 1, \tag{19a}$$

$$\sum_j (1-\delta_{ij})(x_{ji}^{\text{bid}} + x_{ji}^{\text{ask}}) \leq 1. \tag{19b}$$

To retain a purely quadratic QUBO form, the constraints are approximated using squared penalty terms:

$$P_{\text{inflow}} = \lambda_{\text{inflow}} \sum_i \Big(\sum_j (1-\delta_{ij})(x_{ji}^{\text{bid}} + x_{ji}^{\text{ask}}) - 1 \Big)^2, \tag{20a}$$

$$P_{\text{outflow}} = \lambda_{\text{outflow}} \sum_i \Big(\sum_j (1-\delta_{ij})(x_{ij}^{\text{bid}} + x_{ij}^{\text{ask}}) - 1 \Big)^2. \tag{20b}$$

3.6.2 Full QUBO Objective. Combining profit and all penalties gives:

$$\min_{x \in \{0,1\}^n} \Big(-f(x) + P_{\text{flow}} + P_{\text{inflow}} + P_{\text{outflow}} \Big), \tag{21}$$

where

$$f(x) = \sum_{i,j} (1-\delta_{ij}) \big[x_{ij}^{\text{bid}} \log r_{ij}^{\text{bid}} + x_{ij}^{\text{ask}} \log(1/r_{ij}^{\text{ask}}) \big].$$

3.6.3 Formulation Impact on QAOA Ansatz.

Entanglement Strategy. Entanglement structures in QAOA circuits have a critical impact on performance [6]. The entanglement topology — linear, ring, or fully connected — determines how qubits interact and explore the solution space, and tailored schemes can reduce circuit depth while improving results [7, 11]. For example, Carrascal et al. [4] demonstrated that Shifted-Circular-Alternating (SCA) entanglement can yield optimal solutions with a single QAOA layer ($p = 1$), highlighting the benefits of structure-aware designs.

Increased Number of Variables. Incorporating bid and ask rates introduces directional asymmetry to the arbitrage problem, doubling the number of binary variables — and thus qubits — in the QAOA circuit for each currency pair.

Complexity of the Cost Hamiltonian. Bid–ask modeling increases the complexity of the cost Hamiltonian H_C, leading to greater circuit depth and gate count:

- **Additional Linear Terms:** Each bid and ask rate contributes a separate linear term: $\log(r_{ij}^{\text{bid}})$ and $-\log(r_{ij}^{\text{ask}})$.
- **Additional Quadratic Terms:** Constraint penalties (e.g. flow conservation and inflow/outflow) introduce interactions between bid–ask variables, increasing the number of two-qubit gates [15].
- **Greater Circuit Depth:** Enforcing these constraints requires layers of entangling gates; although SCA can distribute them efficiently, the growing number of interactions challenges coherence on NISQ hardware.
- **Increased Resource Demands:** Simulating the expanded Hamiltonian requires more classical memory and computational power, and scaling up may strain both classical simulators and quantum devices.

3.7 Entanglement Strategies

Seven entanglement strategies for QAOA ansätze are presented, each defined by specific CNOT-gate patterns that encode the currency exchange graph and enforce LP constraints (flow conservation, inflow, and outflow).

Shifted Circular Alternating (SCA). entangles qubits in alternating, circular CNOT patterns (e.g., USD → EUR → GBP → USD) to support direct exchanges and triangular cycles.

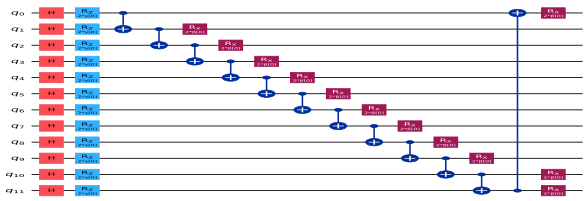

Figure 2: Shifted Circular Alternating (SCA) structure.

High-Coupling Cycle Entanglement (HCEE). extends SCA with skip-2/4/6 and periodic bidirectional CNOTs to model longer cycles and bid-ask spreads. While supporting flow conservation, this dense coupling has the potential to introduce irrelevant correlations between unrelated edges.

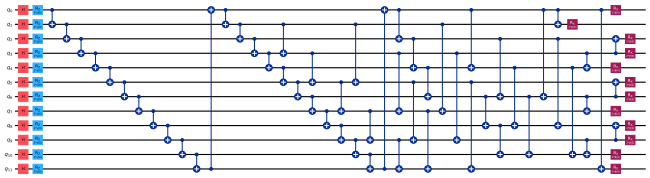

Figure 3: High-Coupling Cycle Entanglement (HCEE).

Reduced-Coupling Exploratory Entanglement (RCEE). simplifies HCEE by omitting skip-6 CNOTs to avoid excessive coupling.

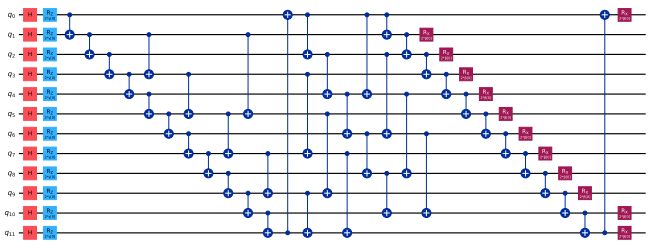

Figure 4: Reduced-Coupling Exploratory Entanglement (RCEE).

Midrange-Distance Exploratory Entanglement (MDEE). uses skip-3 instead of skip-4 connections to capture relevant cycles while avoiding distant couplings.

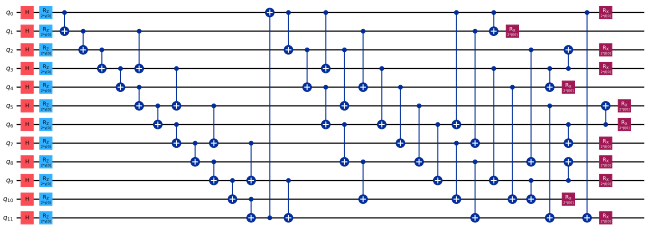

Figure 5: Midrange-Distance Exploratory Entanglement (MDEE).

Same-Bid Reverse Entanglement (SBRE). entangles qubits for reverse bid-bid or ask-ask edges (e.g., x^{bid}_{USDEUR} and x^{bid}_{EURUSD}) to explore pure bid or pure ask cycles.

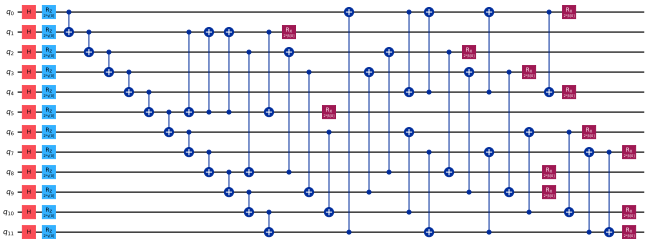

Figure 6: Same-Bid Reverse Entanglement (SBRE).

Bidirectional Symmetric Entanglement Encoding (BSEE). combines two SCA loops in opposite directions with skip-2/4 CNOTs, omitting skip-6 and periodic bidirectional gates to reduce over-entanglement.

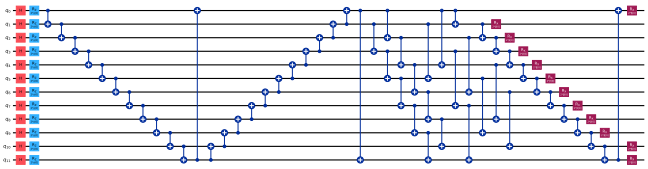

Figure 7: Bidirectional Symmetric Entanglement Encoding (BSEE).

Balanced-Correlation Connectivity Entanglement (BCCE). integrates SCA, skip-2, periodic bidirectional CNOTs, and SBRE to balance specificity and connectivity.

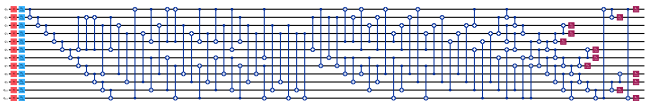

Figure 8: Balanced-Correlation Connectivity Entanglement (BCCE).

Graph-Centric Modular Entanglement (GCME). organizes qubits into modules, each corresponding to one currency. Intra-module circular entanglement enforces local constraints; inter-module bidirectional CNOTs couple reverse edges (e.g., USD→EUR and EUR→USD). This modular, graph-aware design respects LP constraints while avoiding over-entanglement.

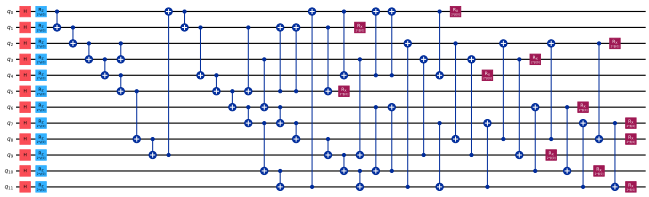

Figure 9: Graph-Centric Modular Entanglement (GCME).

Table 1: Comparison of Entanglement Strategies

Strategy	Linear	Circular	Skip	Bidirectional	Graph-Aware
SCA	Yes	Yes	—	No	No
HCEE	Yes	Yes	2,4,6	Periodic	No
RCEE	Yes	Yes	2,4	No	No
MDEE	Yes	Yes	2,3,6	Periodic	No
SBRE	Yes	Yes	—	Graph-Edges	Yes
BSEE	Yes	Yes	2,4	No	No
BCCE	Yes	Yes	2	Periodic	Yes
GCME	No	Modular	—	Inter-Module	Yes

"Skip" indicates the cycle lengths connected; "Bidirectional" indicates whether bidirectional CNOTs are used; "Graph-Aware" denotes alignment with the currency-graph structure.

4 Experimental Setup

To evaluate performance, the models compared are standard QAOA, QAOA+SCA, and QAOA with each of the seven entanglement structures on FX arbitrage detection tasks. Simulations used Qiskit's `AerSimulator` with a custom noise model to reflect realistic noisy intermediate-scale quantum (NISQ) conditions [13].

Two synthetic FX datasets were used:

- **3-currency network**: Fully connected, bidirectional, with bid and ask rates; 12 qubits total.
- **4-currency network**: Same structure; 24 qubits total.

All noisy simulations used Qiskit's NoiseModel with FakeOsaka simulated backend on an AerSimulator with 8192 shots on 3-currency set and 131,072 shots on the 4-currency set and fixed seeds for reproducibility with COBYLA optimizer [17]; the 3-currency dataset was tested with a 2-layer QAOA using random initialization, while the 4-currency dataset used an 8-layer QAOA with seeded starting points. Based on prior tuning, all QUBO penalty coefficients were set to $\lambda = 100$ — ensuring uniform weighting across flow, inflow, and outflow constraints and preventing any single penalty from dominating despite observed performance variations at other λ values. The optimal objective values for each dataset were found via CPLEX. Each configuration was run five times, and the average

Table 2: Currency Exchange Rate Matrices for 3-Currency and 4-Currency Datasets

3-Currency Bid Rates

	USD	JPY	GBP
USD	1	109.45	0.73
JPY	0.0091	1	0.0067
GBP	1.37	150.32	1

3-Currency Ask Rates

	USD	JPY	GBP
USD	1	109.65	0.74
JPY	0.0092	1	0.0068
GBP	1.38	150.52	1

4-Currency Bid Rates

	USD	JPY	GBP	AUD
USD	1	109.45	0.73	1.34
JPY	0.0091	1	0.0067	0.0123
GBP	1.37	150.32	1	1.83
AUD	0.75	82.21	0.55	1

4-Currency Ask Rates

	USD	JPY	GBP	AUD
USD	1	109.65	0.74	1.35
JPY	0.0092	1	0.0068	0.0124
GBP	1.38	150.52	1	1.85
AUD	0.76	82.41	0.56	1

expectation value for cycle profit and the approximation ratio was recorded, in addition to optimality gap [14].

5 Results

The QAOA results for the 3-currency and 4-currency datasets reveal performance differences across entanglement strategies. Tables 3 and 4 show that entanglement design significantly affects the identification of profitable arbitrage cycles when influenced by noise.

In the 3-currency case, the standard QAOA exhibits limited performance with high variance, likely due to insufficient entanglement depth. The QAOA+SCA variant improves cycle detection but increases variability. GCME achieves near-optimal performance by leveraging its modular structure to target currency clusters while avoiding over-entanglement. SBRE performs similarly, benefiting from graph-specific bid–ask coupling. MDEE also reaches near-optimal solutions with a balanced skip pattern. HCEE, RCEE, BSEE, and BCCE yield moderate gains; denser designs (BSEE, BCCE) may introduce noise-sensitive correlations.

In the 4-currency case, circuit depth, qubit count, and noise increase substantially. The standard QAOA struggles to scale. QAOA+SCA improves robustness, but GCME again delivers near-optimal results. Its inter-module connections adapt well to larger graphs. HCEE and MDEE remain strong, while RCEE and BCCE perform moderately. BSEE and SBRE decline sharply, suggesting their entanglement patterns may be too noise-sensitive for denser graphs.

Table 3: Mean Performance Metrics for 3-Currency Dataset (optimal value of -10.004 found by CPLEX)

Strategy	Expectation	Approx. Ratio	Opt. Gap (%)
QAOA	-6.126 ± 4.752	0.612 ± 0.475	38.76
SCA	-6.372 ± 7.708	0.599 ± 0.771	40.06
HCEE	-7.879 ± 3.643	0.788 ± 0.364	21.24
RCEE	-8.125 ± 3.758	0.812 ± 0.376	18.78
MDEE	-9.875 ± 0.255	0.987 ± 0.026	1.29
SBRE	-7.875 ± 4.254	0.787 ± 0.425	21.28
BSEE	-7.880 ± 3.944	0.788 ± 0.394	21.22
BCCE	-8.002 ± 3.701	0.800 ± 0.370	20.01
GCME	**-10.003 ± 0.001**	**0.999 ± 0.001**	**0.01**

Table 4: Mean Performance Metrics for 4-Currency Dataset (optimal value of -10.574 found by CPLEX)

Strategy	Expectation	Approx. Ratio	Opt. Gap (%)
QAOA	-5.852 ± 4.894	0.553 ± 0.463	44.66
SCA	-6.228 ± 4.868	0.589 ± 0.460	41.10
HCEE	-9.871 ± 0.448	0.934 ± 0.042	6.64
RCEE	-7.977 ± 3.450	0.754 ± 0.326	24.56
MDEE	-10.098 ± 0.689	0.955 ± 0.065	4.50
SBRE	-6.456 ± 4.828	0.611 ± 0.457	38.95
BSEE	-4.468 ± 4.760	0.423 ± 0.450	57.74
BCCE	-8.237 ± 3.522	0.779 ± 0.333	22.10
GCME	**-10.570 ± 0.003**	**0.999 ± 0.001**	**0.04**

GCME consistently produces the most negative expectation values (highest cycle profits) and approximation ratios closest to 1, indicating near-optimal QUBO solutions. Its modular design appears to limit over-entanglement and focus on relevant arbitrage cycles. Circuit depth significantly increased with problem size, but GCME maintained low variance despite deeper circuits in the 4-currency case.

Scalability is critical as currency networks grow. The QUBO formulation requires two binary variables per currency-pair direction (bid and ask), doubling the qubit count and increasing Hamiltonian density. This leads to deeper circuits and higher gate counts. Despite these challenges, GCME scales effectively: its modular design limits entanglement to relevant subgraphs and uses targeted inter-module CNOTs to encode reverse-edge relationships. The results demonstrate that GCME maintains near-optimal performance and low variance even as qubit count increases from 12 to 24, suggesting its suitability for bid-ask arbitrage models under NISQ conditions.

GCME also consistently achieved near-perfect solution quality, with an optimality gap of only 0.01% on both datasets. In practice this means the arbitrage cycles recovered by GCME deliver essentially the same log-return as the exact CPLEX optimum, whereas other strategies showed larger gaps and variance. Highlighting this metric emphasizes final solution quality rather than only intermediate expectation values, underscoring the practical financial relevance of the results. From a scalability perspective, the bid–ask

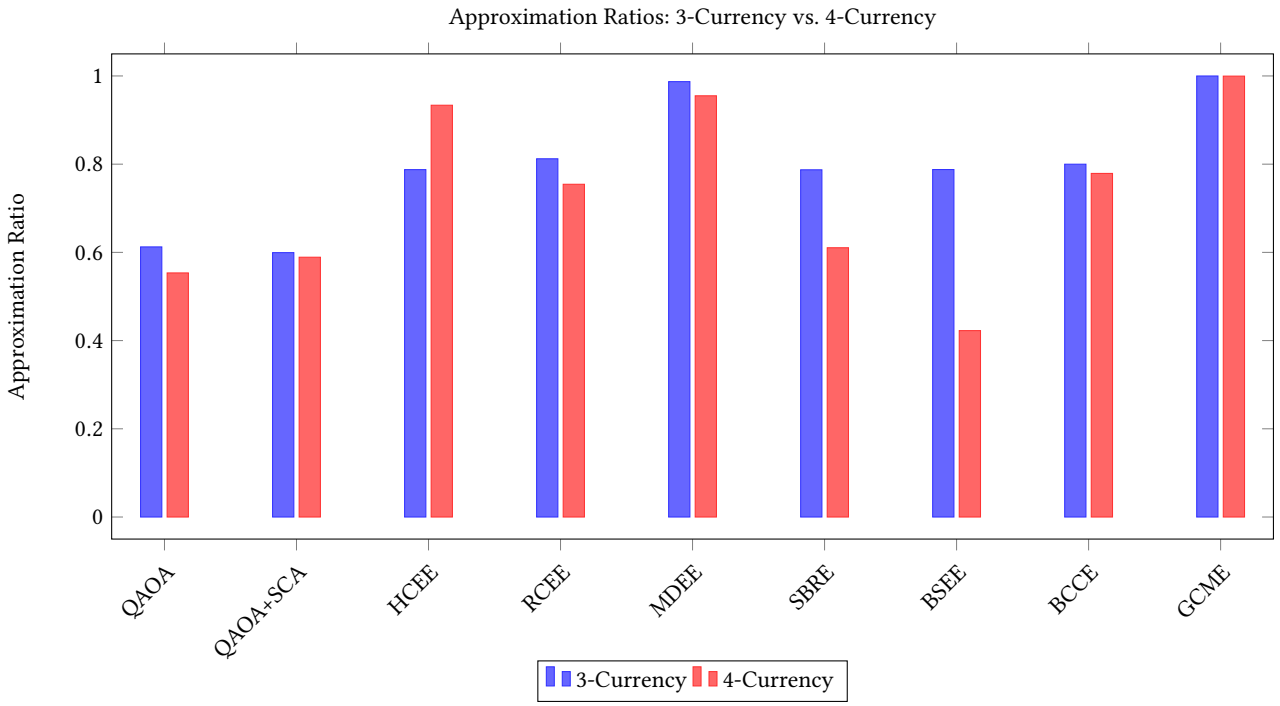

Figure 10: Approximation ratios by entanglement strategy for 3- and 4-currency datasets.

QUBO formulation requires $Q = 2n(n-1) = O(n^2)$ qubits for an n-currency network. Each QAOA layer applies $|E| = O(n^2)$ quadratic cost terms and single-qubit mixers of depth $O(n)$, giving an overall circuit depth scaling of $O(pn^2)$, where p is the number of QAOA layers. In our experiments, moving from three to four currencies doubled the qubit count ($12 \rightarrow 24$) and produced a commensurate depth increase, consistent with this quadratic scaling. These results show that GCME not only maintains near-optimality as problem size grows, but also offers a structured entanglement scheme that mitigates over-correlation, making it well suited to modeling bid–ask arbitrage under NISQ-era constraints.

From a financial standpoint, the results indicate that QAOA with GCME not only maintains stability under noise but also recovers virtually all of the arbitrage profit available in the test networks. For instance, in the 3-currency and 4-currency datasets, the optimality gap was only about 0.01%, meaning that the log-return of the detected cycle differed from the CPLEX optimum by less than one part in ten thousand. In currency trading terms, such a gap corresponds to leaving under one basis point of potential profit unexploited per cycle — a level negligible compared to typical transaction costs or bid–ask spreads. By contrast, other entanglement strategies showed gaps of 20–40%, implying much larger slippage from the true optimum and thus reduced financial relevance. The quadratic scaling of qubits and circuit depth observed here mirrors the combinatorial growth of arbitrage opportunities as the number of currencies increases, with 4 currencies already requiring 24 qubits and significantly deeper circuits than the 12-qubit 3-currency case.

6 Conclusion

This work is a proof-of-concept formulation of currency arbitrage with bid & ask rates as a QUBO and solving it with QAOA under different entanglement strategies. The study focused on 3 and 4-currency networks to demonstrate how circuit design impacts solution quality under NISQ constraints. Across these instances, GCME consistently outperformed other strategies, achieving near-optimal solutions with an optimality gap of only 0.01% and low variance. These results show that modular, graph-aware entanglement can capture the structure of currency exchange networks more effectively than unstructured or overly dense patterns, reinforcing its potential for quantum optimization in finance.

The key direction for future work is to scale GCME-based QAOA to larger FX networks (dozens of currencies) on quantum hardware. Extending beyond simulation to real backends will be essential to evaluate noise resilience and establish the practical feasibility of applying quantum optimization to arbitrage. Additionally, optimizing GCME for real-time processing could enhance its adaptability to time-sensitive optimization problems. These findings pave the way for quantum optimization to address complex financial challenges, opening new avenues for applying quantum computing to practical financial systems, once hardware scales to support more realistic problem sizes.

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Acknowledgments

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